

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 06 · Geometric Sequences

Questions

Student book · questions with writing space

- 22 pages
- Past papers up to January 2026

PURE 2 | TOPIC 06

Geometric Sequences

Your classified route

7

QUESTIONS

55

MARKS

3

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 06 | FAMILY 01

Infinite Sum

WHAT THIS FAMILY ASKS

Recognise the infinite sum demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 State the convergence condition before using the formula.

3 classified questions

24 total marks

6. In a geometric sequence $u_1, u_2, u_3, \dots$

- the common ratio is r
- $u_2 + u_3 = 6$
- $u_4 = 8$

(a) Show that r satisfies

$$3r^2 - 4r - 4 = 0$$

(3)

Adaptive unit 1 · WMA12-01_2022_MayJune_Q06

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6. In a geometric sequence $u_1, u_2, u_3, \dots$

- the common ratio is r
- $u_2 + u_3 = 6$
- $u_4 = 8$

Given that the geometric sequence has a sum to infinity,

(b) find u_1

(3)

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6. In a geometric sequence $u_1, u_2, u_3, \dots$
- the common ratio is r
 - $u_2 + u_3 = 6$
 - $u_4 = 8$

(c) find S_∞

(2)

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4. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Given that, in a particular geometric series,

- the sum of the first three terms is 70.2
- the sum to infinity is 75

find, for this series,

(a) the common ratio,

(4)

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4. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

Given that, in a particular geometric series,

- the sum of the first three terms is 70.2
- the sum to infinity is 75

find, for this series,

(b) the first term. (2)

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- Adaptive unit 1 · WMA12-01_2021_Oct_Q07

(ii) A different geometric sequence has first term a and common ratio r .

Given that

- the second term of the sequence is -6
- the sum to infinity of the series is 25

(a) show that

$$25r^2 - 25r - 6 = 0 \quad (3)$$

(b) Write down the solutions of

$$25r^2 - 25r - 6 = 0 \quad (1)$$

Hence,

(c) state the value of r , giving a reason for your answer, (1)

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(d) find the sum of the first 4 terms of the series.

(2)

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TOPIC 06 | FAMILY 02

Common Ratio

WHAT THIS FAMILY ASKS

Recognise the common ratio demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**1 classified questions**

7 total marks

8. A geometric sequence has first term a and common ratio r

Given that $S_{\infty} = 3a$

(a) show that $r = \frac{2}{3}$

(2)

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8. A geometric sequence has first term a and common ratio r
Given that $S_n = 3a$

Given also that

$$u_2 - u_4 = 16$$

where u_k is the k^{th} term of this sequence,

(b) find the value of S_{10} giving your answer to one decimal place.

(5)

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TOPIC 06 | FAMILY 03

Finite Sum

WHAT THIS FAMILY ASKS

Recognise the finite sum demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 If a sum is given, solve the resulting exponential equation carefully.

3 classified questions

24 total marks

8. (i) A geometric series has first term a and common ratio r .
Prove that the sum of the first n terms of this series S_n is given by

$$S_n = \frac{a(1-r^n)}{1-r} \tag{3}$$

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- (ii) A liquid is to be stored in a barrel.

Due to evaporation, the volume of the liquid in the barrel at the end of each year is 8% less than the volume of the liquid in the barrel at the start of the year.

At the start of the first year, the barrel is filled with 150 litres of the liquid.

- (a) Show that the amount of the liquid in the barrel at the end of 6 years is approximately 91 litres.

(2)

At the start of each year a new barrel is filled with 150 litres of the liquid so that, at the end of 40 years, there are 40 barrels containing the liquid.

- (b) Calculate the total amount of the liquid, to the nearest litre, in the 40 barrels at the end of 40 years.

(3)

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7. A geometric sequence has first term a and common ratio r , where $r > 0$

Given that

- the 3rd term is 20
- the 5th term is 12.8

(a) show that $r = 0.8$

(1)

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7. A geometric sequence has first term a and common ratio r , where $r > 0$
- Given that
- the 3rd term is 20
 - the 5th term is 12.8

(b) Hence find the value of a .

(2)

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7. A geometric sequence has first term a and common ratio r , where $r > 0$
- Given that
- the 3rd term is 20
 - the 5th term is 12.8

Given that the sum of the first n terms of this sequence is greater than 156

(c) find the smallest possible value of n .

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(4)

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6.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

A software developer released an app to download.

The numbers of downloads of the app each month, in thousands, for the first three months after the app was released were

$$2k - 15 \quad k \quad k + 4$$

where k is a constant.

Given that the numbers of downloads each month are modelled as a geometric series,

(a) show that $k^2 - 7k - 60 = 0$

(2)

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(4)

6. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

A software developer released an app to download.
The numbers of downloads of the app each month, in thousands, for the first three months after the app was released were

$$2k - 15 \quad k \quad k + 4$$

where k is a constant.
Given that the numbers of downloads each month are modelled as a geometric series,

The **total** number of all downloads of the app is predicted to exceed 3 million for the first time in the N th month.

(c) Calculate the value of N according to the model.

(3)

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